ALP-ttbar Toy MC v1.6: Post-registration Results for a Magnetic-Field and Material-Budget Scan of the Two-Hypothesis Vertex Fit for Displaced Light ALP → e^+e^-

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Preregistration (frozen and public before any v1.6 result): doi:10.5281/zenodo.23942338, github.com/wspaik5/ALP_ttbar_Toy_MC_v1.6_Preregistration_Field_Material_Scan (README SHA-256 a9e4b64b...8835c4). Previous versions: v1.5 doi:10.5281/zenodo.23924862, v1.4 doi:10.5281/zenodo.23880375

Disclaimer. This is an independent, simplified, numpy-only toy study. It is not affiliated with the authors of Refs. [1, 2] or with the CMS, LHCb or ALICE Collaborations or the FCC study, uses parametrised kinematics and assumed detector numbers, and should be read as qualitative trends only. It makes no claim about the design or performance of a real detector.

Summary. v1.5 found no gain from a two-hypothesis vertex fit at 17–50 MeV in a 3.8 T field. v1.6 repeats the v1.5 analysis in eight preregistered configurations of the field B (3.8 → 0.2 T) and the material scale factor f (1 → 0.25). **Material is the effective handle:** the 17 MeV loss in reach from conversions falls from 127× to 65× (f = 0.5) and 35× (f = 0.25), faster than the conversion yield alone. **Lowering the field does not help at 17 MeV**: the conversion vertex resolution degrades from 7.9 to 48 mm, the material veto is abandoned for B ≤ 1 T and the loss rises to 184–188×. The $\Delta\chi^2$ separation itself improves strongly at low field (at 200 MeV without mass window 130× → 52×), but the 1/B loss of momentum resolution offsets it once the mass window is used; the best field at 100–200 MeV is about 1 T. Four of seven predictions are confirmed (P1, P2, P3, P7) and three refuted (P4, P5, P6).

1. Question and preregistration

Alimena et al. [1] extended the top-associated long-lived ALP search of Rygaard et al. [2] to a → e^+e^- at generator level; this toy series studies detector-level effects not included there. In v1.5 a free vertex absorbs a small ALP opening $\Delta\varphi$ by a shift $s_0 = \Delta\varphi/\Delta\omega$, leaving a mismatch $\Delta\varphi^2/(2\Delta\omega)$ that scales as 1/B, while conversions and bremsstrahlung [3] scale with the material budget. **Q1:** how does the conversion-induced loss at 17–200 MeV ($c\tau$ = 1 mm) change with B and f? **Q2:** how do the $\Delta\chi^2$ separation, the conversion vertex resolution and the curvature resolution change? Configurations, scalings, a sanity gate, validations Y1–Y3 and predictions P1–P7 were published before any v1.6 computation. Field and material optimisation [8, 9], reduced-field running [10] and field-based conversion rejection (φ_V) [11] are established; the claim is limited to this search.

2. Method (as preregistered)

Unchanged from v1.5: mini simulation, samples and seeds, H_{conv} (= v1.4 fit (b) [5, 6]), H_{ALP}, $\Delta\chi^2$ and the scan_v15 selection with its V-nomass variant. **Configurations:** R (3.8 T, f = 1), B2, B1, B05, B02 (2.0, 1.0, 0.5, 0.2 T), M05, M025 (3.8 T, f = 0.5, 0.25) and C (2.0 T, f = 0.5). **Scalings:** curvature with B; pair-mass momentum resolution 0.02 × 3.8 T/B; angular scattering term × √f; all X_0 × f (conversion probability, scattering, bremsstrahlung). **Statistics:** first 100,000 base conversion pairs and first 60,000 base signal pairs per mass (whole chunks), weights rescaled. **Sanity gate** (passed): config R reproduces the v1.5 vertex radius and $\Delta\chi^2$ for 100.000% of pairs in all five samples; fit success 99.3–99.99% for H_{conv} and H_{ALP} in all 40 configuration-samples; no $\Delta\chi^2 < -0.1$. The median reported vertex resolution grows from 7.3 mm (R) to 40.7 mm (B02) for conversions, as expected.

3. Validation (independent samples only)

Y1 (null, no brems). In v1.5 the preregistered χ^2_1 reference (Wilks' theorem [12]) failed, because with a free vertex the opening angle is identified only at second order and the null lies on a parameter boundary; the correct reference is $½\chi^2_0 + ½\chi^2_1$ [13, 14] (v1.5; record in Appendix A). v1.6 used this boundary reference from the start: fraction with $\Delta\chi^2 < 10^{-3}$ 0.448 (0.2 T) and 0.508 (2.0 T), median of the non-zero part 0.512 and 0.463 — pass. This confirms at two further fields, independently of v1.5, that the boundary mixture and not χ^2_1 is the right null distribution for this test. **Y2** (2 mrad opening, no brems): median $\Delta\chi^2$ 0.045, 0.20, 0.87, 2.53, 6.81 at 3.8, 2.0, 1.0, 0.5, 0.2 T, increasing monotonically as B decreases — pass (the 3.8 T value reproduces v1.5 X2). **Y3** (low-field propagation): hits agree with the analytic circle to 3 × 10^{-8} µm at 0.2 T — pass.

4. Results

Fig. 10 v1.6 (preregistered, doi:10.5281/zenodo.23942338): field and material scan (toy MC, subsample)

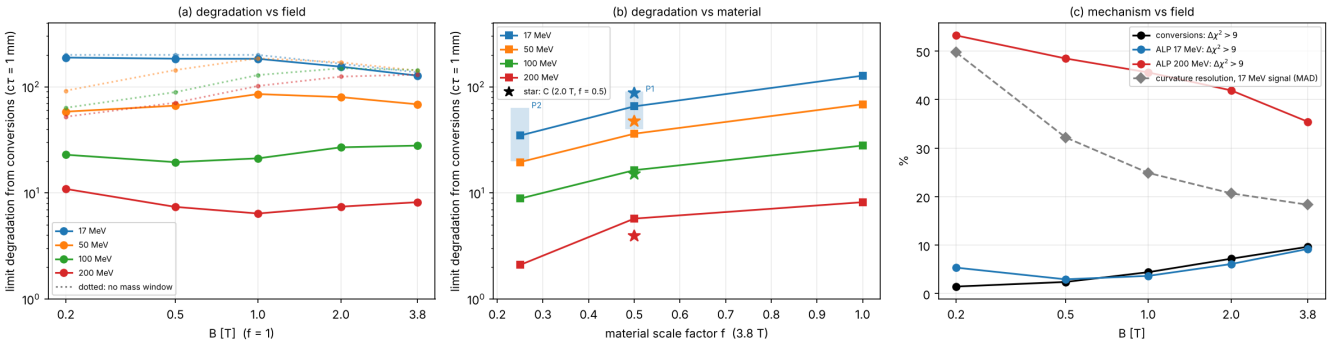


Figure 10. (a) Degradation versus field at f = 1 (dotted: without mass window). (b) Degradation versus material at 3.8 T, with the preregistered ranges P1, P2 and the combined point C (stars). (c) Fraction of pairs with $\Delta\chi^2 > 9$ for conversions and for 17 and 200 MeV ALP pairs, and the core curvature resolution of 17 MeV signal tracks, versus field.

config	B, f	17 MeV	50 MeV	100 MeV	200 MeV	no mass window 17 / 200	conv. core σ
R	3.8 T, 1	127	68	28	8.2	137 / 130	7.9 mm
B2	2.0 T, 1	155	80	27	7.4	165 / 125	10.3 mm
B1	1.0 T, 1	184	85	21	**6.4**	200 / 102	14.6 mm
B05	0.5 T, 1	184	66	20	7.4	200 / 71	22.9 mm
B02	0.2 T, 1	188	58	23	10.9	200 / **52**	47.9 mm
M05	3.8 T, 0.5	**65**	36	16	5.7	70 / 70	4.9 mm
M025	3.8 T, 0.25	**35**	20	8.8	2.1	37 / 36	3.7 mm
C	2.0 T, 0.5	87	48	15	3.9	92 / 68	6.9 mm

Table 1. Limit degradation from conversions (×), cτ = 1 mm, v1.5 two-hypothesis selection, subsample. Config R agrees with the full-statistics v1.5 values (126, 68, 28, 9.1). For B ≤ 1 T at 17 MeV the scan chooses no material veto at all, so the value is that of an unvetoed selection.

config	conv. $\Delta\chi^2 > 9$	ALP 17 / 50 / 100 / 200 MeV $\Delta\chi^2 > 9$	curvature resolution 17 / 200 MeV
R	9.6%	9.2 / 10.4 / 16.9 / 35.4%	18.3 / 22.5%
B2	7.2%	6.1 / 9.1 / 23.6 / 41.9%	20.7 / 27.7%
B1	4.4%	3.6 / 12.7 / 32.8 / 45.6%	24.9 / 37.1%
B05	2.4%	2.9 / 21.8 / 39.3 / 48.5%	32.2 / 53.4%
B02	1.4%	5.3 / 30.7 / 44.7 / 53.2%	49.7 / 89.2%
M05	4.2%	4.2 / 4.7 / 12.1 / 31.9%	11.5 / 14.5%
M025	1.7%	1.8 / 2.2 / 9.8 / 31.3%	7.9 / 10.0%

Table 2. Mechanism (Q2). Curvature resolution: MAD of $\omega_{fit}/\omega_{true} - 1$ for signal tracks in the H_{ALP} fit.

Preregistered predictions vs. outcome

prediction (frozen)	outcome	verdict
P1: M05, 17 MeV in 40–90×	65.4×	confirmed
P2: M025, 17 MeV in 20–63×	34.7×	confirmed
P3: B02, 17 MeV worse than R	188.4× vs 127.3×	confirmed
P4: B02, 200 MeV better than R	10.9× vs 8.2×	**refuted**
P5: 17 MeV ALP $\Delta\chi^2 > 9$ rises monotonically R → B02 to ≥ 30%; conversions lower at B02	9.2 → 6.1 → 3.6 → 2.9 → 5.3%; conv. 9.6 → 1.4%	**refuted** (ALP part)
P6: curvature resolution ≥ 10× worse at B02 than R	49.7% vs 18.3% (2.7×)	**refuted**
P7: C at 17 MeV between M05 and B2	86.6× (65.4×, 154.6×)	confirmed

5. Interpretation

Material. Less material reduces conversions and bremsstrahlung together: the conversion vertex core improves (7.9 → 3.7 mm), fewer conversions look opening-angle-like (9.6% → 1.7%), and the 17 MeV loss falls roughly in proportion to f, faster than the √f expected from the yield alone. **Field at 17 MeV.** As B decreases the two conversion tracks bend less, the tangent point along the track is poorly constrained and the vertex resolution grows to ≈ 48 mm, so the material veto stops working (it is not used at all for B ≤ 1 T). The larger mismatch does not compensate: at 17 MeV the ALP opening (≈ 1–2 mrad) still leaves only ≈ 20–40 µm at 0.2 T, comparable to the scattering and bremsstrahlung noise, so the fraction of 17 MeV pairs with large $\Delta\chi^2$ even falls (P5). **Field at 50–200 MeV.** The $\Delta\chi^2$ separation improves as expected (Y2; Table 2) and, without a mass window, the loss at 200 MeV falls from 130× to 52×. With the mass window the improvement is offset by the 1/B momentum resolution, which widens the mass peak; the optimum is near 1 T (P4 refuted). Since the mass and $\Delta\chi^2$ are independent in this toy, the no-mass-window numbers isolate the vertex information. **Curvature.** At 3.8 T the curvature resolution of electrons (18%) is already set by bremsstrahlung, so lowering the field degrades it only 2.7× (P6 refuted). Within this toy: reducing material helps at all masses; reducing the field trades conversion-vertex location against opening-angle visibility and is not favourable for the lightest ALPs.

6. Deviations from preregistration

(1) After the fits of configurations R, B02 and M05 had finished, their analysis was run once as an interim look while the remaining configurations were still being fitted; nothing in the code, selection or predictions was changed afterwards. **(2)** Code: *mag.py*, *bkg.py* and *minitrack.py* read B and f from environment variables (defaults reproduce v1.5; identity gate 100.000%); *fit2h.py* additionally stores the H_{ALP} curvatures for P6. **(3)** Y2 regenerates the v1.5 X2 sample sequence (0 mrad, then 2 mrad) so that the 3.8 T value reproduces v1.5; Y3 compares with the analytic circle through the origin. **(4)** At 0.2 T one of 8,000 Y1 pairs failed to fit (7,999 used). No other changes.

7. Limitations

All v1.0–v1.5 limitations apply (transverse plane, barrel only, idealised layers, no pattern recognition, toy material map, crude counting rule, selection optimised on the same samples, truth used for the material list, cosh η and the scattering momentum). The field is uniform and the geometry fixed; low-field points are not a model of any real or forward detector. All material is scaled by one factor; the 1/B momentum-resolution scaling is an assumption; charge identification and two-track separation at low field are not modelled. Only the transverse opening is used. Subsampled statistics. Only trends and ratios are meaningful; realistic answers need the experiments' tools [7].

8. Future directions

(i) A three-dimensional vertex fit including the r–z opening, which the field does not bend and bremsstrahlung barely changes, with the single-Gaussian brems model; (ii) the same with a full Gaussian-sum filter [5, 6] (ACTS); (iii) a dimuon control study on the same simulation; (iv) extension to $\mu^+\mu^-$ above 200 MeV and other long-lived-particle signatures (e.g. $B \to K^{(*)}a$, searched for by LHCb in dimuons [16], which would need a forward geometry); (v) a location-dependent material model instead of a single scale factor; (vi) **calibrated test statistics**: before assuming χ^2_k (Wilks [12]) for a new hypothesis test, check its regularity conditions in the preregistration, and make the null reference a selectable part of the code (Wilks χ^2_k; boundary mixtures following Chernoff [13] and, for several boundary or correlated parameters, Self and Liang [14]; reparametrisation where a parameter enters only at higher order [15]; or the empirical distribution of a dedicated null sample of pseudo-experiments). Boundary-aware asymptotics of this kind are already standard in collider searches [17]; the change proposed here concerns the procedure of this series.

Appendix A. Record from v1.5: why Wilks' theorem failed for the two-hypothesis test

This appendix reproduces, for completeness, the v1.5 finding (doi:10.5281/zenodo.23924862, Sec. 4) on which the v1.6 validation Y1 is based. No number in it was recomputed for v1.6. **What was preregistered.** The v1.5 null test X1 (zero opening, no bremsstrahlung, 8000 pairs) expected $\Delta\chi^2 = \chi^2_{conv} - \chi^2_{ALP}$ to follow χ^2_1, as Wilks' theorem [12] predicts for one extra parameter: median 0.30–0.65 and 3–8% above 3.84. **What was observed.** Median 0.000 and 2.15% above 3.84, so X1 failed. Nothing in the code or selection was changed. **Why Wilks' theorem does not apply.** The theorem requires that the tested value lies inside the parameter space and that the parameter changes the prediction at first order (non-singular information matrix). With a free vertex both conditions fail for the opening angle Δφ: the vertex shift $s_0 = \Delta\varphi/\Delta\omega$ absorbs Δφ at first order, so the hits respond only to $\psi \propto \Delta\varphi^2 \geq 0$. A brute-force profile scan (opening fixed at 0.1–5 mrad, all other parameters re-fitted) showed that this is not a minimiser problem and that the χ^2 rise grows as $\Delta\varphi^4$ instead of $\Delta\varphi^2$ (log–log slope 4.3; singular information matrix [15]). The null $\psi = 0$ lies on the boundary of the parameter space, where the likelihood-ratio statistic follows $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ [13, 14].

quantity (v1.5 X1 sample)	observed	½χ^2_0 + ½χ^2_1 (boundary)	χ^2_1 (preregistered)
fraction with $\Delta\chi^2 = 0$	51.3%	50%	0%
median of the non-zero part	0.457	0.455	–
fraction > 2.71	4.88%	5.0%	10%
fraction > 3.84	2.15%	2.5%	5%

Table A1. v1.5 quantitative check of the boundary mixture.

Fig. 9 Why validation X1 failed: the opening angle is identified only at second order (post-hoc diagnosis)

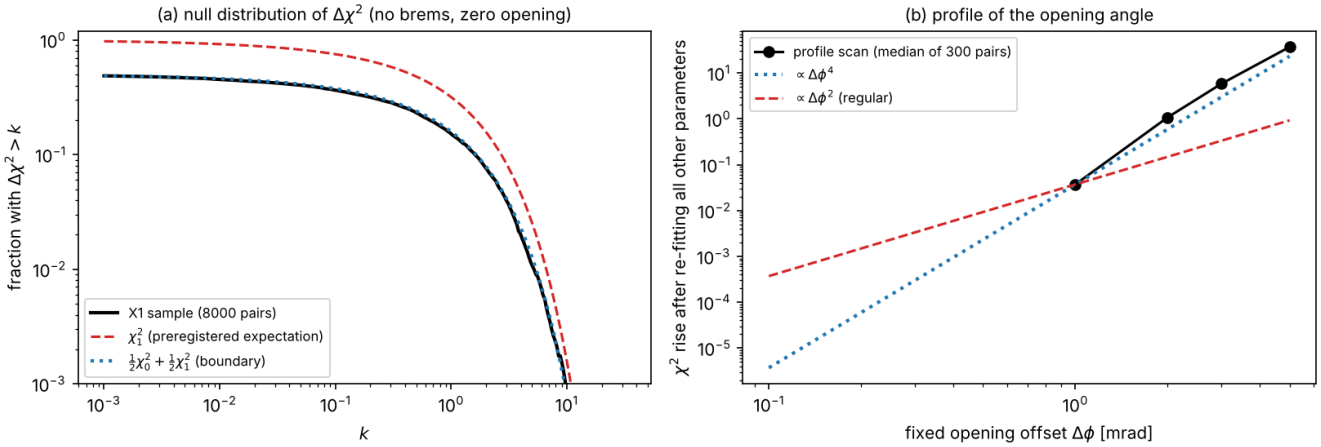


Figure 9 (reproduced from v1.5, post-hoc). (a) Null distribution of $\Delta\chi^2$ on the X1 sample compared with χ^2_1 and the boundary mixture. (b) χ^2 rise when the opening is fixed and all other parameters are re-fitted, with $\Delta\varphi^2$ and $\Delta\varphi^4$ references.

Consequences carried into v1.6. The v1.5 main result was unaffected because the exemption threshold was scanned, not derived from χ^2_1. The test remains usable with the mixture as reference, so the v1.6 preregistration fixed ½χ^2_0 + ½χ^2_1 as the Y1 reference in advance; Y1 passed at 0.2 T and 2.0 T (Sec. 3), confirming the v1.5 diagnosis at two further fields. The v1.5 X1 failure itself stays recorded as a failed preregistered validation; it was not re-framed.

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Links

This record: doi:10.5281/zenodo.23952888 · code: github.com/wspaik5/ALP_ttbar_Toy_MC_v1.6_Postregistration_Field_Material_Scan
Preregistration: doi:10.5281/zenodo.23942338 · github.com/wspaik5/ALP_ttbar_Toy_MC_v1.6_Preregistration_Field_Material_Scan